Graphic Era

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B.Tech ECE VII
Name of the Paper: Telecommunication Switching
Paper Code: TEC-702

Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty)
(iv) Each sub-question carries 10 marks.

Q1. (2X10=20 Marks)

a. Describe Manual Switching system in detail? CO1

b. Explain the working principle of a simple telephone communication system, highlighting the key components involved and their roles in establishing a connection. CO1

c. Explain the concept of crossbar switching, including its architecture and advantages over earlier switching systems like the Strowger switch. CO1

Q2. (2X10=20 Marks)

a. Explain the architecture and working of centralized and distributed stored program control (SPC) in electronic space division switching. How do these approaches impact the system's performance and scalability? CO1

b. Explain the processes of sampling, quantization, and encoding in time division switching. How are these concepts fundamental to digital signal transmission? CO

c. Describe the operation of basic time division space switching and basic time division time switching. Highlight the differences between the two methods. CO

Q3. (2X10=20 Marks)

a. Explain the concept of **Grade of Service (GoS)** in network traffic engineering and its significance in maintaining optimal network performance. Include examples of parameters used to measure GoS. CO

b. A telecommunication system receives an average of **200 calls per hour**, and the average call duration is **3 minutes**. Calculate the **traffic intensity (in Erlangs)** and determine if a system with 15 circuits can handle this load with a blocking probability of less than 2%. CO

c. Describe the **SS7 signaling protocol** used in telecommunication networks. What are its key components, and how does it ensure reliability and security in signaling? CO

Q4. (2X10=20 Marks)

a. Explain the concept of Integrated Digital Networks (IDNs). Discuss how they facilitate the seamless integration of various communication systems and their significance in modern digital communication. CO

b. Compare and contrast Time Division Multiplexing (TDM) and Frequency Division Multiplexing (FDM). Discuss their working principles, advantages, and applications in digital communication systems. CO

c. Define Integrated Services Digital Network (ISDN) and explain its primary objectives. Discuss its architecture and the services it provides to users. CO

Q5. (2X10=20 Marks)

a. Compare and contrast the different switching techniques (circuit switching, packet switching, and message switching). Provide examples of where each technique is commonly used. CO

b. Explain the OSI model's seven layers. How do link-to-link and end-to-end layers ensure reliable communication in data networks? CO

c. What are the principles of Asynchronous Transfer Mode (ATM) networks? How do they support data CO

communication across different types of networks?